

DROWNING IN THE CHILDREN'S EMERGENCY

CONTENTS

1. DEFINITION	2
2. RISK FACTORS	2
3. PATHOPHYSIOLOGY OF DROWNING	2
4. COMPLICATIONS OF DROWNING	3
5. ACUTE MANAGEMENT	3
5.1 FOCUSED HISTORY	3
5.2 MANAGEMENT	3
6. THERMOREGULATION	5
6.1 PHYSIOLOGIC EFFECTS OF HYPOTHERMIA	5
6.2 METHODS OF REWARMING	5
6.3 REWARMING	5
7. INVESTIGATIONS	6
7.1 BIOCHEMICAL / POCT	6
7.2 RADIOLOGICAL	6
8. PROGNOSTIC FACTORS	6
9. DISPOSITION	7
9.1 ADMISSION	7
9.2 DISCHARGE	7
10. REFERENCES	9

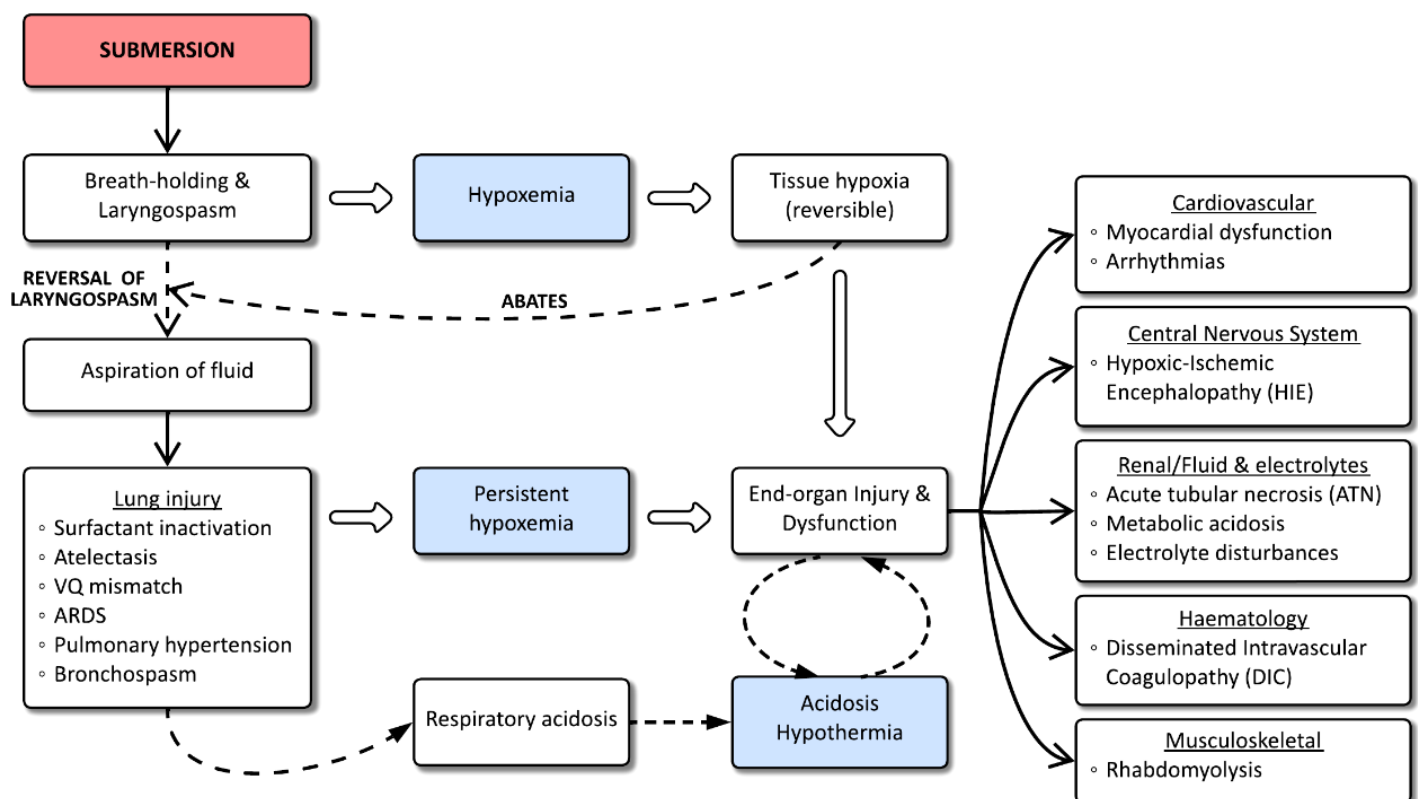
1. DEFINITION

- Drowning is a process resulting in primary respiratory impairment from submersion/immersion in a liquid medium resulting in asphyxiation (suffocation) ⁽¹⁾.
- Drowning does not occur without aspiration
- Terms like “wet”, “dry”, “near-drowning”, “non-fatal drowning” should be abandoned

2. RISK FACTORS

- Epilepsy
- Cardiac arrhythmias (channelopathies)
- Alcohol & illicit drug use (adolescents)

3. PATHOPHYSIOLOGY OF DROWNING



- The main pathophysiological mechanism is hypoxemia 2' lung injury
- Resultant tissue hypoxia, systemic acidosis ± hypothermia result in end-organ dysfunction & injury

4. COMPLICATIONS OF DROWNING

SYSTEM	COMPLICATIONS	SYSTEM	COMPLICATIONS
RESP	- Surfactant inactivation - Atelectasis - VQ mismatch - ARDS - Pulmonary hypertension - Bronchospasm	METABOLIC / ELECTROLYTES	- Hypothermia - Respiratory acidosis - Metabolic acidosis - Hyper/hyponatremia
		RENAL	Acute tubular necrosis
CVS	- Myocardial dysfunction - Arrhythmias - Hypotension	HAEM	- Hemolysis - DIVC
CNS	- Cerebral edema - Seizures	MSK	Rhabdomyolysis

5. ACUTE MANAGEMENT

If no life-threatening conditions identified, priorities of management:

1. Reverse hypoxemia/hypoxia
2. Thermoregulation
3. Neuroprotection

5.1 FOCUSED HISTORY

- Cause of drowning?
- Duration of downtime (submersion + CPR)
- Was CPR performed? (Bystander or paramedics)

5.2 MANAGEMENT

MX STEP	DETAILS
Resuscitation	As per PALS algorithms (cardiac arrest, dysrhythmias)
Airway / Breathing	Restore oxygenation to reverse hypoxemia - Consider intubation for PEEP if persistent T1RF on high FiO₂ - Increase PEEP by 1 – 2 cmH₂O (max. 10 cmH₂O) while watching BP (reduced venous return 2' ↑ intrathoracic pressure)
	Consistent minute ventilation to achieve stable pCO ₂ levels - Volume control - Cuffed ETT for intubation to achieve closed system
	NGT insertion to decompress stomach → reduce risk of further lung injury from aspiration of gastric contents

MX STEP	DETAILS													
Circulation	Reverse shock - Fluid bolus with <u>20 ml/kg</u> of crystalloids - Consider early inotropic support for myocardial dysfunction Treat reversible causes: - Electrolytes Hypocalcemia, hypomagnesemia, hypokalemia - Acidosis Correct with restoration of adequate oxygenation, ventilation, circulatory volume expansion. NaHCO₃ is usually unnecessary - Temperature Hypothermia (aim temperature ~ 33 – 35 °C) Effects of hypoxia & acidosis on heart most commonly: - Tachycardia → Bradycardia → PEA → Asystole - Ventricular fibrillation (VF) is not commonly seen → suspect underlying LQTS, brugada, or intoxication (e.g., TCA overdose)													
Disability	Start neuroprotective measures for patients with significant hypoxic injury (i.e., prolonged downtime, required CPR, GCS < 14) Goal is to prevent secondary CNS insult from disordered homeostasis - Raised ICP 2' cerebral edema or noxious stimuli (e.g., pain) - Acidosis - Seizures - Increased metabolic demands (e.g., fever, hypoglycemia) - Release of excitatory neurotransmitters <table border="1"> <tr> <td rowspan="6">Neuroprotective measures</td><td>Head up 30°</td></tr> <tr> <td>Target low normal pCO₂ (35 – 40 mmHg) with stable minute ventilation</td></tr> <tr> <td>High normal BP targets</td></tr> <tr> <td>High normal Na levels (145 – 150 mmol/L)</td></tr> <tr> <td>- <u>4 ml/kg</u> of 3% NaCl (raises serum Na by 3 mmol/L)</td></tr> <tr> <td>Reduce metabolic demands</td></tr> <tr> <td></td><td> - Target normothermia - Control fever - Normoglycemia </td></tr> <tr> <td></td><td>Reduce noxious stimuli</td></tr> <tr> <td></td><td> - Adequate analgesia for pain - Keep well sedated - IV lignocaine <u>1 – 2 mg/kg</u> (200mg), 2 – 3 mins before intubation or suctioning </td></tr> </table>	Neuroprotective measures	Head up 30°	Target low normal pCO ₂ (35 – 40 mmHg) with stable minute ventilation	High normal BP targets	High normal Na levels (145 – 150 mmol/L)	- <u>4 ml/kg</u> of 3% NaCl (raises serum Na by 3 mmol/L)	Reduce metabolic demands		- Target normothermia - Control fever - Normoglycemia		Reduce noxious stimuli		- Adequate analgesia for pain - Keep well sedated - IV lignocaine <u>1 – 2 mg/kg</u> (200mg), 2 – 3 mins before intubation or suctioning
Neuroprotective measures	Head up 30°													
	Target low normal pCO ₂ (35 – 40 mmHg) with stable minute ventilation													
	High normal BP targets													
	High normal Na levels (145 – 150 mmol/L)													
	- <u>4 ml/kg</u> of 3% NaCl (raises serum Na by 3 mmol/L)													
	Reduce metabolic demands													
	- Target normothermia - Control fever - Normoglycemia													
	Reduce noxious stimuli													
	- Adequate analgesia for pain - Keep well sedated - IV lignocaine <u>1 – 2 mg/kg</u> (200mg), 2 – 3 mins before intubation or suctioning													
Exposure	Remove wet clothing & avoid over-exposure Rectal (core) temperature should be taken with a thermometer that can read < 35°C Control & prevent fever													
Others	Empiric antibiotics for aspiration pneumonia - IV Augmentin <u>30 mg/kg</u> (1.2g) per dose													

6. THERMOREGULATION

- Hypothermia causing cold-induced arrhythmias and myocardial dysfunction is associated with cold water (< 10°C) drowning, rare in local context
- It is usually 2' excessive heat loss from over-exposure and evaporation.
- Measure rectal temperature with a probe that can read below 35°C

6.1 PHYSIOLOGIC EFFECTS OF HYPOTHERMIA

SEVERITY	TEMPERATURE	REWARMING	EFFECTS
Mild	≥ 32°C to < 35°C	Passive	- Depressed mental status, shivering
Moderate	≥ 28°C to < 32°C	Active	- A state of unresponsiveness below 30°C. - Cyanosis, tissue edema and rigidity with shallow respiratory efforts and pulses. - Atrial fibrillation and dysrhythmias.
Severe	< 28°C	Active	- Patient may appear dead with undetectable vital signs with dilated & unresponsive pupils. - Ventricular fibrillation, asystole, or extreme bradycardia.

6.2 METHODS OF REWARMING

METHOD	DETAILS	
Passive	Warmed blankets, remove wet clothing	
Active (external)	To chest, axillae, and groin (avoid peripheries) Heated air blower (e.g., Bair hugger), radiant heat, chemical heat packs	
Active (internal)	Non-invasive	Heated, humidified O ₂ (41°C to 45°C) Warmed saline (40°C to 44°C)
	Invasive	Lavage of pleura, bladder, stomach & peritoneum with warmed saline (39°C to 41°C) Extracorporeal rewarming (ECMO or cardiac bypass)

6.3 REWARMING

- Avoid aggressive rewarming as hyperthermia worsens cerebral injury
- Rate of rewarming for patients with suspected hypoxic ischemic encephalopathy
 - [*hemodynamically unstable*] 1 – 2°C per hour ⁽²⁾ ➔ Target = 35°C – 36°C
 - [*hemodynamically stable*] 0.5°C per hour ➔ Target = 35°C – 36°C
- Patients with a full GCS and shivering from hypothermia ➔ Target = 36°C – 37°C

7. INVESTIGATIONS

All drowning patients should have a CXR, and ECG performed

7.1 BIOCHEMICAL / POCT

- 12-lead ECG
- Blood gas with micro-electrolytes
- Capillary blood glucose
- FBC, PT/PTT/INR
- Renal Panel 2, Magnesium, ionized Calcium
- Liver function tests (LFT), Creatinine kinase (CK), Troponin I, CKMB
- Lactate (if clinical features of shock)
- Consider: Urinary drug screen, Ethanol levels (if suspicious circumstances or presentation)

7.2 RADIOLOGICAL

- Chest X-Ray
- Consider CT Brain if patient's GCS is ≤ 13

8. PROGNOSTIC FACTORS

- No robust review that exists as numbers are small
- Patients with ≥ 3 of the following indicators would have a 5% probability of survival. Patients with ≤ 2 of the following indicators are associated with a 90% chance of full and complete recovery ⁽³⁾.
 1. Submersion time > 5 minutes
 2. Age < 3 years
 3. Delay in initial resuscitative efforts by > 10 minutes
 4. Coma on arrival to hospital
 5. Initial pH < 7.1
- Patients with persistent profound coma (flaccid or posturing) 2 – 6 hours after drowning will likely have brain death or moderate to severe neurological impairment; Those who show definite signs of improvement will fare better with normal to near-normal recovery. ⁽⁴⁾
- Other indicators of poor outcome ⁽⁵⁾:
 - CPR > 25 minutes
 - Submersion > 25 minutes
 - On-going CPR on arrival to ED

9. DISPOSITION

- If patient was resuscitated with return of spontaneous circulation (ROSC), start post-cardiac arrest care to minimize further physiological insults to end-organs. (Refer to ANNEX A)

9.1 ADMISSION

9.1.1 PICU/HD

- Child received CPR
- Requires oxygen supplementation ≥ 5 L/min
- Requires mechanical ventilation
- Documented acidemia on blood gas ($\text{pH} < 7.2$)
- GCS < 14

9.1.2 GENERAL WARD

- Requires supplementary oxygen < 5 L/min
- Chest radiograph abnormal
- Abnormal lung findings & respiratory vital signs

9.2 DISCHARGE

- Discharge from CE after 6 – 8H of monitoring if
 - Normal vital signs and GCS levels
 - Normal chest radiograph
 - Normal physical examination findings

ANNEX A: POST-CARDIAC ARREST CARE CHECKLIST

AIRWAY/BREATHING (OXYGENATION & VENTILATION)	
Target normal saturations, SpO ₂ 94 – 99%	☐
Target low normal pCO ₂ levels (35 – 40 mmHg); Employ use of ETCO ₂ monitoring for overall trends.	☐
Prevent swings & fluctuations in pCO ₂ levels (i.e., stable minute ventilation) ▫ Ventilator mode: VC mode for patients with a cuffed ETT ▫ Optimizing patient-ventilator dyssynchrony by ensuring: ▪ Ventilator settings and support is appropriate for patient's age/weight. ▪ Keep well sedated with enough analgesia, KIV paralyze patient if persistent dyssynchrony despite optimizing sedation & eliminating other causes.	☐
CIRCULATION (HEMODYNAMIC MONITORING)	
Continuous cardiac monitoring	☐
Frequent BP monitoring (every 30 minutes), targeting BP > 5 th percentile for age and sex	☐
Guide therapy with markers of cardiac output: ▫ Lactate ▫ Titrate with fluid boluses ± inotropes / vasopressors ▫ Urine output ≥ 0.5 ml/kg/h	☐
Optimize electrolytes (K ≥ 4.0 mmol/L, iCa ≥ 1.2 mmol/L, Mg ≥ 1.0 mmol/L)	☐
DISABILITY (NEUROPROTECTION & SEDATION)	
Provide adequate sedation with analgesia and anxiolytics	☐
Neuroprotective measures ▫ Nurse with head up at 40° ▫ pCO₂ & blood pressure targets (as above.) ▫ Reduce metabolic demands (thermoregulation) ▪ Temperature targets (see below.) ▪ Maintain homeostasis (metabolic control) ▫ Reduce noxious stimuli: ▪ Keep well sedated (analgesia and anxiolytic); consider paralysis ▪ IV lignocaine prior to suctioning	☐
EXPOSURE (TEMPERATURE TARGETS)	
If patient is comatose, target core temperature 33°C – 36°C	☐
If patient is shivering or hemodynamically unstable, target core temperature 36°C – 37.5°C	☐
Prevent & treat fever	☐

10. REFERENCES

1. *Recommended guidelines for uniform reporting of data from drowning: the "Utstein style"*. **Idris AH, Berg RA, Bierens J, et al. , 59(1):45-57.** 1, 2003, Resuscitation, Vol. 59.
2. *Immersion, near-drowning and drowning*. **Golden FS, Tipton MJ, Scott RC.** 2, 1997, Br J Anaesth, Vol. 79, pp. 214 - 225.
3. *Prognostic factors in pediatric cases of drowning and near-drowning*. **JP., Orlowski.** 5, 1979, JACEP., Vol. 8, pp. 176 - 179.
4. *Drowning, near-drowning, and ice-water submersions*. **JP., Orlowski.** 1, 1987, Pediatr Clin North Am., Vol. 34, pp. 75 - 92.
5. *Pediatric submersions: prehospital predictors of outcome*. **Quan L, Kinder D.** 6, 1992, Pediatrics., Vol. 90, pp. 909 - 913.